Replication Report of Marsh, Vachon, & Jones (2008).

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Introduction

Original citation

Marsh, J. E., Vachon, F., & Jones, D. M. (2008). When does between-sequence phonological similarity promote irrelevant sound disruption? *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 34, 243-248.

Original study

In their original paper, Marsh Vachon, and Jones (2008) examined the effect of between-sequence phonological similarity on serial recall and free recall. It is well known that serial recall of a list of target items (words, letters, or numbers) is markedly disrupted by auditory distractors. Within the modular working memory model (Baddeley & Hitch, 1974), this so-called irrelevant sound effect has been attributed to phonological interference (Salamé & Baddeley, 1982). More specifically, it was postulated that irrelevant speech has obligatory access to the phonological-loop component of working memory, where it overwrites the to-be remembered target representation that are maintained in a phonological code. One implication of this model is that between-stream similarity—that is, the phonological similarity between the target items and the auditory distractors—should increase interference (Salamé & Baddeley, 1982). However, contrary to these predictions, most previous studies (Jones & Macken, 1995; Larsen, Baddeley, & Andrade, 2000; LeCompte & Shaibe, 1997) have found that between-stream similarity has no effect on serial recall (but see Eagan & Chein, 2012). Marsh et al. have replicated this null finding. In addition, they showed that free recall, in contrast to serial recall, was affected by between-stream phonological similarity. This finding is in contrast to the predictions of the modular working memory model (Salamé & Baddeley, 1982). However, it is to-be expected if people use phonological similarity to generate plausible list candidates in free

recall. Assuming that category cuing is more important for free recall than for serial recall, the interference-by-process account put forward by Marsh and colleagues (2008) predicts between-sequence phonological similarity effects on free recall, but no such effects on serial recall.

Target of replication

In this replication, as part of the Reproducibility Project: Psychology, we sought to reproduce Marsh et al.'s (2008) finding that there was a "pronounced disruptive effect of between-sequence phonological similarity on the free recall of lists of end rhyming words." Specifically, the original authors hypothesized that a "between-sequence similarity effect [would] occur for the free but not serial recall of categorized material that is defined not semantically but phonologically." They found evidence of an interaction between instruction type and sound condition that supports this hypothesis.

Methods

Power Analysis

G*Power (Faul, Erdfelder, Lang, & Buchner, 2007) was used to calculate the power analysis. An a priori power analysis showed that a sample of 118 participants was necessary to detect an interaction between instruction type (free recall, serial recall) and sound condition (quiet, phonologically dissimilar, phonologically similar) of the size $\eta_p^2 = 0.06$ (computed from the *F*-value provided in the target article) with a statistical power of $1 - \beta = .95$, given that $\alpha = .05$.

Materials

The original use of materials below was followed precisely, unless changes were necessary because German participants (instead of UK participants, as in the original study) were tested, as specified below.

> "The rhyme-generating program Rhymer (WriteExpress; http://www.rhymer.com) was used to generate words by entering 30 words each with a different but common end rhyme (e.g., "ide" as in "side"). The monosyllabic end rhymes generated for each word were then chosen

> and scored for frequency with the Thorndike–Lorge word count (Thorndike & Lorge, 1944) as were the originally entered words. For each end rhyme, the resultant six highest frequency words were chosen as the TBI [to-be ignored] auditory items, whereas the 10 lowest frequency words were chosen as the TBR [to-be remembered] visual items.
>
> This method of constructing and selecting materials differed from previous studies (e.g., Beaman, 2004) for the following reasons. Items chosen in these prior studies were derived from category norms: Although the common attribute of an end rhyme means that end-rhyme words form categories, we were unaware of any "end-rhyme category norms." This is unfortunate to the extent that the dominance relations between the items cannot be determined: We cannot be sure that the highest frequency items are the first items participants will generate when presented with the end rhyme as a cue. Logically however, there should be a correlation between frequency of use in the English language and dominance level, and because of this, the highest frequency items were chosen as the irrelevant items. Limits imposed by restricted category sizes forced omissions of items not scored for frequency in the Thorndike–Lorge word count, and the removal of homophones resulted in the use of six words as TBI items and 10 words as the TBR items in each list.
>
> The items within the TBR and TBI sequences were presented in a randomized order that was the same for each participant. We recorded the TBI items in a male voice at an even pitch at a sampling rate of 44.1 kHz in 16-bit resolution using Sound Forge 5.0 (Sonic Inc., Madison, WI). Each TBI item was then digitally edited to a duration of 500 ms. We presented the TBI items stereophonically over headphones at approximately 65–70 db(A) using SuperLab Pro software (Cedrus Corp., San Pedro, CA)."

Given that the experiment was replicated with German participants, it was necessary to select German words as stimulus material. Different rhyme-generating programs (http://www.echtreim.de/, http://www.reimbuch.net/) were used to select German rhyme words. Pseudowords were removed from the list. Furthermore, names (including city names), idioms, or foreign words were not selected unless commonly used and known in Germany. The frequency of the words was determined using a database of German word frequencies (http://wortschatz.uni-leipzig.de). Finally, a different type of software was used to record and play the sounds, and to run the experiment.

Procedure

The original procedure below was followed precisely:

“The 30 TBR lists and 30 irrelevant sound sequences were divided into two 15-list sets. For each instruction type, half the participants were randomly assigned to one of the two 15-list sets.

Each 15-list set was further subdivided into five blocks of three lists. Within each three-list block, the TBR lists were assigned randomly to one of the three sound conditions. The order of the three sound conditions within each three-trial block was counterbalanced across participants such that an equal number of participants were assigned to each of the six possible order permutations.

When a TBR list was assigned to the phonologically similar condition, the sound sequence corresponding to the category represented by that list was presented. When a TBR list was assigned to the phonologically dissimilar condition, the sound sequence was randomly chosen from the 15 categories not represented by the TBR list.

Participants were seated at a viewing distance of approximately 60 cm from a PC monitor on which TBR items were displayed in a central position. Lists of TBR items appeared in lowercase black 72-point Times New Roman font one word at a time against a white background. Each word—whether TBR or TBI—was presented for 500 ms with no interstimulus interval. The end of the list was notified by the appearance of a red “WAIT” sign that was displayed for 9 s before a green “RECALL” sign was shown to indicate the end of the retention period and start of the retrieval period. Each of the TBI items was presented three times in the retention period.

Participants were seated in individual cubicles. They were informed that they would be presented with fifteen ten-word lists and that each list would be presented one word at a time on the computer monitor from which they were asked to memorize as many words as possible and write the words they remembered down on their recall sheets when a “RECALL” cue appeared on the screen.

On presentation of the RECALL cue, participants given free recall instructions were instructed to try and remember as many of the words in any order, whereas participants given serial recall instructions were instructed to remember the words according to their original order of presentation. Participants from the serial recall group were also told that they could leave gaps in their recall protocols if necessary. Participants in this group were also instructed to guess the original position if they had an item available to them for recall but could not remember its position. Recall sheets contained 15 columns of 10 rows each: Participants given serial recall instructions were provided with specially prepared recall sheets with serial positions marked on them; the free-recall instructed group had the same recall sheets but without the serial positions marked.

Participants were informed that they would have 30 s to retrieve as much as they could of the list and that after this time a tone would sound to signal the beginning of the next list

(some 5 s later). Participants were instructed to ignore any sound that they heard through the headphones and were told that they would not be tested on its content at any point in the experiment."

Analysis

A 2 (instruction type: free vs. serial recall) x 3 (sound condition: quiet, phonologically similar, phonologically dissimilar) mixed ANOVA was run to replicate the analysis that indicated phonologically similar irrelevant items were more disruptive than phonologically dissimilar items in free recall, but not in serial recall.

As in the original study, different scoring criteria were used to analyze serial recall data and free recall data before entering the data into this analysis:

> "The raw serial recall data were scored according to a strict serial recall criterion—recalled items were only scored as correct if they corresponded to their presentation position—whereas a free recall criterion was employed to score the raw free recall data: Recalled items were scored as correct regardless of serial position."

Proportion correct was defined as the number of items that were correctly recalled, divided by the number of items that were presented.

Differences from Original Study

The original procedure was followed as closely as possible, but German words were used as stimulus material instead of English words (because a sample of German participants was tested instead of a sample of UK participants). Thus, new sets of target words and distractor words were generated based on German word norms. Different software was used to record and play the distractor stimuli, and to run the experiment. Furthermore, the students in the original study participated for course credit while the participants in the replication study were students from different academic disciplines who participated for monetary compensation (see Korn & Hogan, 1992, for a discussion of how different types of incentives may affect the willingness to participate in an experiment). Finally, confidence ratings were not assessed. In the original study, on completion of the experiment participants were instructed to return to the beginning of the recall sheets, and give confidence ratings for their answers. The confidence ratings were considered

irrelevant to the replication study because analyses of these confidence ratings were not reported in the original paper. Given that the confidence ratings were collected after the original study was completed, this aspect of the procedure could not have any influence on the main results.

Primary Replication

Actual Sample

One-hundred twenty one students from Heinrich-Heine University participated for monetary compensation. Data of three participants could not be analyzed. One data file was accidentally overwritten, and two participants failed to fill out the recall sheets properly[1]. Analyses are based on the remaining 118 participants (77 female, 41 male; mean age = 22.79, *SD* = 3.75). All participants were fluent German speakers (116 participants were native German speakers).

Differences From Pre-Data Collection Methods Plan

None.

Results

Data preparation

Participants' answers were transcribed, and compared against the target items of the to-be remembered lists. The transcripts were double-checked. Spelling errors were ignored.

Correct Responses

Figure 1 shows the proportion of correct recall in free and serial recall as a function of irrelevant sound condition. As in the original study, phonologically similar distractors disrupted recall compared with phonologically dissimilar items in free recall, $t(58) = 4.42$,

[1] One participant thought that the recall test would start at the end of the experiment, after all lists had been presented, resulting in a failure to remember any words at all. The other participant did not recall list 1, and wrote the list 2 items into the column reserved for list 1 items, the list 3 items into the column reserved for list 2 items, and so on.

$p < .001$, $\eta_p^2 = .25$, but not in serial recall, $t(58) = 1.35$, $p = .18$, $\eta_p^2 = .03$, but, in contrast to the original study, the global interaction between instruction type and sound condition was not significant in a 2 × 3 mixed analysis of variance (ANOVA) with instruction type (free vs. serial recall) as a between-participants factor and sound condition (quiet, phonologically dissimilar, phonologically similar) as a within-participant factor, $F(2,232) = 1.63$, $p = .20$, $\eta_p^2 = .01$.

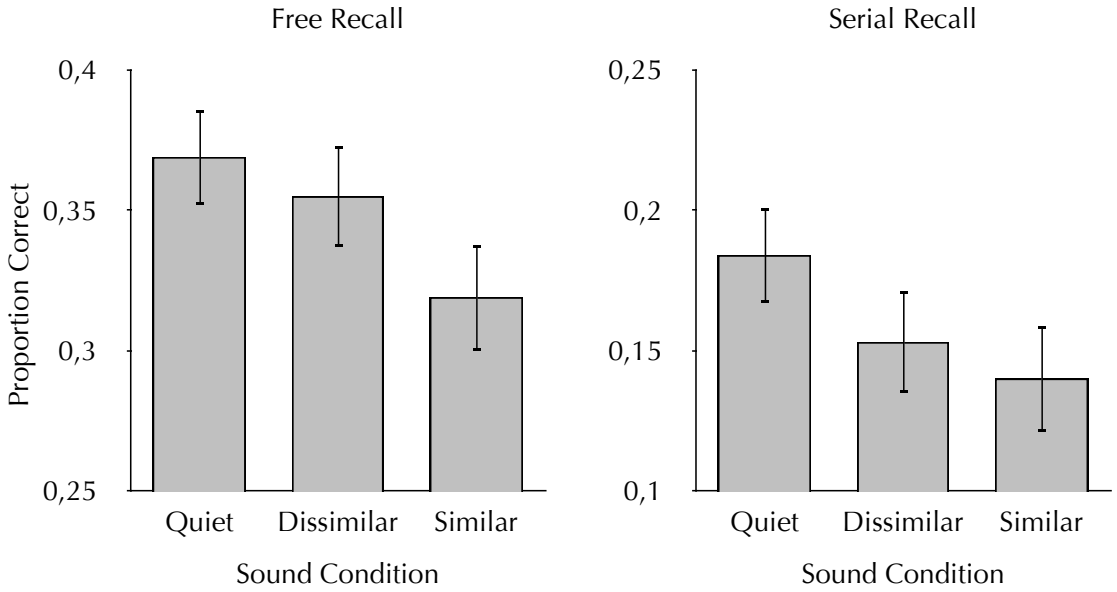


Figure 1: Proportion of correct recall in free and serial recall as a function of irrelevant sound condition (quiet, phonologically dissimilar, similar). Error bars represent .95 confidence intervals.

Intrusions

As in the original study, intrusion errors were analyzed in a supplementary analysis. Two types of intrusions were distinguished. Related intrusions are responses that matched one of the six high-frequency items presented as distractors in the phonologically dissimilar condition. Other-item intrusions refer to items that were not presented at all during the experiment. Figure 2 shows the mean total number of intrusions as a function of instruction type (free recall, serial recall), intrusion type (related, unrelated), and sound condition (quiet, phonologically dissimilar, phonologically similar).

Of particular importance is the interaction between intrusion type and sound condition. Marsh et al. (2008) reported that there was a significant interaction between intrusion type and sound condition performed on the mean number of intrusions in free recall. This pattern can be replicated here, $F(2,116) = 15.28$, $p < .001$, $\eta_p^2 = .21$, indicating that phonologically similar distractors intruded into memory. In contrast to the original study, the interaction between intrusion type and sound condition was significant in the serial recall group, too, $F(2,116) = 24.56$, $p < .001$, $\eta_p^2 = .30$. As in the original study, the three-way interaction between instruction type (free recall, serial recall), intrusion type (related, unrelated), and sound condition (quiet, phonologically dissimilar, phonologically similar) was not significant.

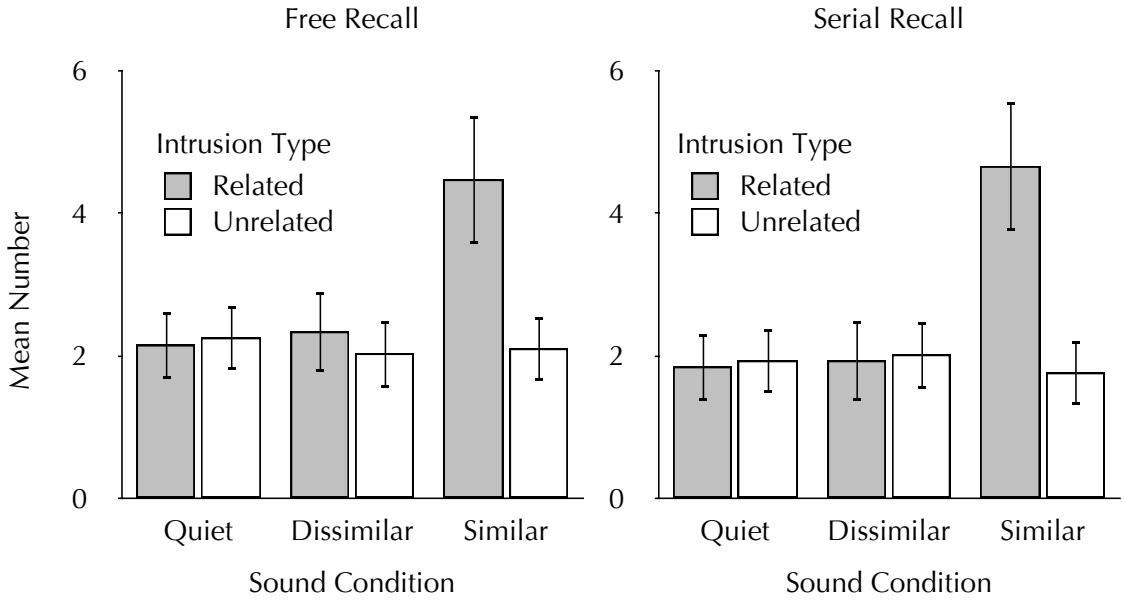


Figure 2: Mean number of related- and unrelated-item intrusions in free recall and serial recall as a function of irrelevant sound condition (quiet, phonologically dissimilar, phonologically similar). Error bars represent .95 confidence intervals.

Discussion

There are many studies in the irrelevant sound literature (Jones & Macken, 1995; Larsen et al., 2000; LeCompte & Shaibe, 1997) showing that between-stream phonological similarity has no effect on serial recall. Marsh et al. (2008) replicated this finding, and

showed that free recall is more disrupted by phonologically similar in comparison to phonologically dissimilar distractors. They summarized their findings as follows:

> In the present study, we established for the first time a between-sequence phonological similarity effect using free recall as the focal task and replicated the finding that such an effect does not occur in the context of serial recall (cf. Jones & Macken, 1995).

This pattern of findings was replicated in the present study. In agreement with previous findings (Jones & Macken, 1995; Larsen et al., 2000; LeCompte & Shaibe, 1997; Marsh et al., 2008) showing that between-stream phonological similarity does not affect serial recall, between-stream phonological similarity had no effect on performance in the serial recall task, but free recall was more disrupted by phonologically similar distractors in comparison to phonologically dissimilar distractors. However, in contrast to the original study, the critical interaction between instruction type (free recall, serial recall) and sound condition (quiet, phonologically dissimilar, phonologically similar) was not significant.

A supplementary analysis confirmed Marsh et al.'s finding that between-stream phonological similarity leads to intrusions errors in free recall. In the present study, phonologically similar speech led to more related-item intrusions in serial recall, too, suggesting that serial recall performance is not completely immune to the effects of between-stream phonological similarity (see Eagan & Chein, 2012, for a similar finding). Given that this effect was absent in the original study, further research is necessary before clear conclusions can be drawn about this issue.

There were some differences between the present study and the original study that could have affected the results. To replicate the experiment with German participants, it was necessary to generate German stimulus material using German rhyme databases, and German frequency norms. It is possible to speculate that the German words used in the present study and the English words used in the original study may have differed in a number of variables, such as word frequency, valence, and so on. Furthermore, the correlation between frequency of use and dominance level (see Methods section) may differ between German and English words. Finally, the phonological-similarity effect on free recall is likely to depend on the use of specific memory strategies (category cuing). It

is possible that populations differ in the likelihood or efficacy with which these strategies are used.

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